Detection of Cyber Harassers on Social Media

CSD 330
Security Analytics

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CSD 330 – Security Analytics

Problem Statement

The goal is to classify a message as bullying or not, to help make the internet a safer place.

Use Cases

Virtually unlimited. Bullying can happen over almost any social media platform or any online forum.

Motivation

Social media has opened doors for people who misuse it to undermine, harass, humiliate, threaten and bully others.

There is a lack of resources to detect and hinder its occurrence.

On average, about 16% students have admitted that they have been bullied at some point in their lives, as per surveys conducted in the United States.

The problem is hence indeed grave and needs to be addressed.

Data Set

“Detecting Insults in Social Commentary”

The dataset I have used for this project is the one provided at Kaggle, which was part of a contest held 8 years ago.

Insult	Date	Comment
1	201206181	"You fuck your dad."
0	201205281	"i really don't understand your point.\xa0 It seems that you are mixing apples and oranges."
0		"A\\xc2\\xa0majority of Canadians can and has been wrong before now and will be again.\\n\\nUnless you're supportive of the idea that nothing is
0		"listen if you dont wanna get married to a man or a women DONT DO IT. what would it bother you if gay people got married stay in your lane do yo
0	20120619(	"C\xe1c b\u1ea1n xu\u1ed1ng \u0111\u01b0\u1eddng bi\u1ec3u t\xecnh 2011 c\xf3 \xf4n ho\xe0 kh\xf4ng ? \nC\xe1c ng\u01b0 d\xe2n ng\u1ec
0	201206201	"@SDL OK, but I would hope they'd sign him to a one-year contract to start with. Give him the chance to be reliable and productive, but give themse
0	20120503(	"Yeah and where are you now?"
1		"shut the fuck up. you and the rest of your faggot friends should be burned at the stake"
1	201205021	"Either you are fake or extremely stupid...maybe both..."
1	201206201	"That you are an idiot who understands neither taxation nor women's health."
0	20120620(	"@jdstorm dont wish him injury but it happened on its OWN and i DOUBT he's injured, he looked embarrassed to me"
0	20120530(	"Be careful,Jimbo.OG has a fork with your name on it."
0		"@tonnyb Or they just don't pay attention "
0		"Hmmm. Perhaps some who are too pig-faced to get laid and therefore have zero chance of getting pregnant from such activity hold something aga
0		"@barrettmarson Huh? Her income was $21,912. Her spending was $5,470. My math puts that at 24.9%."
1	20120611(	"FOR SOME REASON U SOUND RETARDED. LOL. DAMN. WHERE U BEEN NEGRO"
1	201203201	"You with the 'racist' screen name\n\nYou are a PieceOfShit....."
0		"Oh! & cheating as well...1966 cup was a cheat fest, & you still dare to post that comment..shame on you lol"
1	20120320(	"your such a dickhead..."
1	20120320	"Your a retard go post your head up your #%&*"
0	20120612(	"@EephusBlue\xa0Makes you want to say "Mike MacWHOgal?""
0	20120611	"http://www.youtube.com/watch?v=tLYLLPHKRU4"
0	20120502	"And you know they've burned holes in all the carpeting."
0	20120503(	"you are a land creature. You would drown...."
0		"To all those doom-and-gloomers, the chicken-littles who claim the sky is falling and that society is collapsing...Canada has had gay marriage nation
0	201206121	"Craig, we have been saying from the start that this is a very entertaining show.\xa0 It awakens the horny little Goth in me.\xa0 I met my boyfriend
0	20120529(	"Strange you are talking about that when there were Chelsea fans in here belittling the Community Shield"
0	20120612(	"But how would you actually get the key out?"

https://www.kaggle.com/c/detecting-insults-in-social-commentary.

Requirements

- sklearn
- imblearn
- pandas
- seaborn
- matplotlib
- nltk
- re
- xgboost

Use pip to install

Preprocessing

- Removal of special characters using regular expressions.
- Removal of extra whitespace.
- Change to lower case.
- Tokenisation
- Stop word removal
- Lemmatisation
- Stemming
- Removal of null values

Classifiers Used

Initially, the following classifiers were used:

- Naïve Bayes
- Decision Tree
- Support Vector Classifier
- Random Forest
- Logistic Regression
- K-Nearest Neighbours

Metrics for each are shown in later slides.

Naïve Bayes

The metrics are as follows

Confusion Matrix

```
[[1299  655]
 [ 289  404]]
```

Classification Report

	precision	recall	f1-score	support
0	0.82	0.66	0.73	1954
1	0.38	0.58	0.46	693
accuracy			0.64	2647
macro avg	0.60	0.62	0.60	2647
weighted avg	0.70	0.64	0.66	2647

from sklearn.naive_bayes import GaussianNB

Decision Tree

The metrics are as follows

Confusion Matrix

```
[[1878   76]
 [ 493  200]]
```

Classification Report

	precision	recall	f1-score	support
0	0.79	0.96	0.87	1954
1	0.72	0.29	0.41	693
accuracy			0.79	2647
macro avg	0.76	0.62	0.64	2647
weighted avg	0.77	0.79	0.75	2647

from sklearn.tree import DecisionTreeClassifier

Support Vector Classifier (SVC)

The metrics are as follows

Confusion Matrix

```
[[1895   59]
 [ 442  251]]
```

Classification Report

	precision	recall	f1-score	support
0	0.81	0.97	0.88	1954
1	0.81	0.36	0.50	693
accuracy			0.81	2647
macro avg	0.81	0.67	0.69	2647
weighted avg	0.81	0.81	0.78	2647

from sklearn.svm import SVC

Random Forest

The metrics are as follows

Confusion Matrix

```
[[1778  176]
 [ 320  373]]
```

Classification Report

	precision	recall	f1-score	support
Negative	0.85	0.91	0.88	1954
Positive	0.68	0.54	0.60	693
accuracy			0.81	2647
macro avg	0.76	0.72	0.74	2647
weighted avg	0.80	0.81	0.81	2647

from sklearn.ensemble import RandomForestClassifier

Logistic Regression

The metrics are as follows

Confusion Matrix

```
[[1829  125]
 [ 333  360]]
```

Classification Report

	precision	recall	f1-score	support
Negative	0.85	0.94	0.89	1954
Positive	0.74	0.52	0.61	693
accuracy			0.83	2647
macro avg	0.79	0.73	0.75	2647
weighted avg	0.82	0.83	0.82	2647

```
from sklearn.linear_model import LogisticRegression
```

K-Nearest Neighbours

The metrics are as follows

Confusion Matrix

```
[[1776  178]
 [ 419  274]]
```

Classification Report

	precision	recall	f1-score	support
Negative	0.81	0.91	0.86	1954
Positive	0.61	0.40	0.48	693
accuracy			0.77	2647
macro avg	0.71	0.65	0.67	2647
weighted avg	0.76	0.77	0.76	2647

from sklearn.neighbors import KNeighborsClassifier

Hyperparameter Tuning

Done using GridSearchCV.

This was only done for the following three classifiers that performed the best.

We will only deal with these from now on.

Support Vector Classifier

Random Forest

Logistic Regression

```
from sklearn.model_selection import GridSearchCV
```

Support Vector Classifier using GridSearch

Parameter	Definition	Values
C	Regularisation Parameter.	[0.1,1, 10, 100]
gamma	Kernel Coefficient.	[1,0.1,0.01,0.001]
kernel	Kernel Type	['rbf', 'poly', 'sigmoid']

The metrics are as follows:

Confusion Matrix

```
[[1836  118]
 [ 339  354]]
```

Classification Report

	precision	recall	f1-score	support
0	0.84	0.94	0.89	1954
1	0.75	0.51	0.61	693
accuracy			0.83	2647
macro avg	0.80	0.73	0.75	2647
weighted avg	0.82	0.83	0.82	2647

```
from sklearn.svm import SVC
```

Random Forest using GridSearch

Parameter	Definition	Values
n-estimators	Number of trees	[10, 100, 1000]
max_features	Number of features	['sqrt', 'log2']

The metrics are as follows:

Confusion Matrix

```
[[1783  171]
 [ 316  377]]
```

Classification Report

	precision	recall	f1-score	support
0	0.85	0.91	0.88	1954
1	0.69	0.54	0.61	693
accuracy			0.82	2647
macro avg	0.77	0.73	0.74	2647
weighted avg	0.81	0.82	0.81	2647

```
from sklearn.ensemble import RandomForestClassifier
```

Logistic Regression using GridSearch

Parameter	Definition	Values
C	Inverse of regularization strength	[100,10,1.0,0.1,0.01,0.001]
solver	Algorithm	['newton-cg', 'lbfgs', 'liblinear','sag','saga']
penalty	Norm used in penalization.	['none', 'l1', 'l2','elasticnet']

The metrics are as follows:

Confusion Matrix

```
[[1821  133]
 [ 332  361]]
```

Classification Report

	precision	recall	f1-score	support
0	0.85	0.93	0.89	1954
1	0.73	0.52	0.61	693
accuracy			0.82	2647
macro avg	0.79	0.73	0.75	2647
weighted avg	0.82	0.82	0.81	2647

```
from sklearn.linear_model import LogisticRegression
```

XGBoost

I tried using XGBoost independently, as well as using GridSearch, but the metrics did not improve and hence this approach was trashed.

Confusion Matrix

```
[[1866   88]
 [ 414  279]]
```

Classification Report

	precision	recall	f1-score	support
0	0.82	0.95	0.88	1954
1	0.76	0.40	0.53	693
accuracy			0.81	2647
macro avg	0.79	0.68	0.70	2647
weighted avg	0.80	0.81	0.79	2647

```
import xgboost as xgb
```

Sampling

Since the dataset is quite imbalanced (1:3), I try to look at a combination of under sampling and oversampling on the training set to improve my classifier. For this purpose, I used **SMOTETomek** on the training data.

Using this updated training data, I re-ran the optimal classifiers (as returned by GridSearch)

```
from imblearn.combine import SMOTETomek
```

Support Vector Classifier using GridSearch after sampling

Parameter	Definition	Values
C	Regularisation Parameter.	[0.1,1, 10, 100]
gamma	Kernel Coefficient.	[1,0.1,0.01,0.001]
kernel	Kernel Type	['rbf', 'poly', 'sigmoid']

The metrics are as follows:

Confusion Matrix

```
[[1424  530]
 [ 238  455]]
```

Classification Report

	precision	recall	f1-score	support
0	0.86	0.73	0.79	1954
1	0.46	0.66	0.54	693
accuracy			0.71	2647
macro avg	0.66	0.69	0.66	2647
weighted avg	0.75	0.71	0.72	2647

from sklearn.svm import SVC

Random Forest using GridSearch after sampling

Parameter	Definition	Values
n-estimators	Number of trees	[10, 100, 1000]
max_features	Number of features	['sqrt', 'log2']

The metrics are as follows:

Confusion Matrix

```
[[1541  413]
 [ 328  365]]
```

Classification Report

	precision	recall	f1-score	support
0	0.82	0.79	0.81	1954
1	0.47	0.53	0.50	693
accuracy			0.72	2647
macro avg	0.65	0.66	0.65	2647
weighted avg	0.73	0.72	0.73	2647

```
from sklearn.ensemble import RandomForestClassifier
```

Logistic Regression using GridSearch

Parameter	Definition	Values
C	Inverse of regularization strength	[100,10,1.0,0.1,0.01,0.001]
solver	Algorithm	['newton-cg', 'lbfgs', 'liblinear','sag','saga']
penalty	Norm used in penalization.	['none', 'l1', 'l2','elasticnet']

The metrics are as follows:

Confusion Matrix

```
[[1476  478]
 [ 236  457]]
```

Classification Report

	precision	recall	f1-score	support
0	0.86	0.76	0.81	1954
1	0.49	0.66	0.56	693
accuracy			0.73	2647
macro avg	0.68	0.71	0.68	2647
weighted avg	0.76	0.73	0.74	2647

```
from sklearn.linear_model import LogisticRegression
```

Majority Vote Classifier

In the hope of getting a better result, I used a majority vote classifier with hard and soft voting.

However, there was no improvement.

Hard Voting

Confusion Matrix

```
[[1653  301]
 [ 371  322]]
```

Classification Report

	precision	recall	f1-score	support
0	0.82	0.85	0.83	1954
1	0.52	0.46	0.49	693
accuracy			0.75	2647
macro avg	0.67	0.66	0.66	2647
weighted avg	0.74	0.75	0.74	2647

```
from sklearn.ensemble import VotingClassifier
```

Soft Voting

Confusion Matrix

```
[[1686  268]
 [ 320  373]]
```

Classification Report

	precision	recall	f1-score	support
0	0.84	0.86	0.85	1954
1	0.58	0.54	0.56	693
accuracy			0.78	2647
macro avg	0.71	0.70	0.71	2647
weighted avg	0.77	0.78	0.77	2647

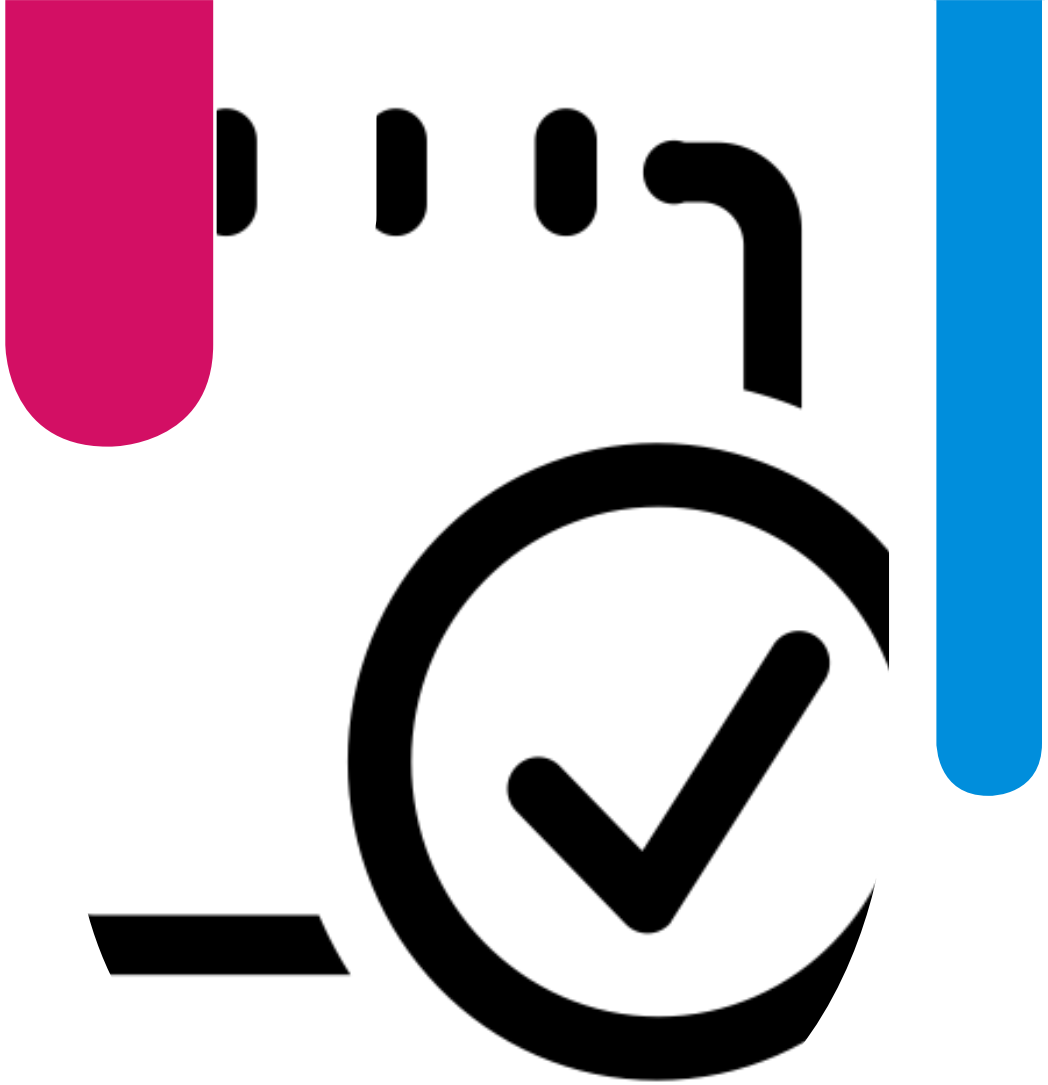
Observations

The metric most important to us is **recall** on the positive class. This is because a false negative is far more harmful than a false negative.

Hence, we prefer accuracy over recall.

XGBoost and VotingClassifier did not help us, and GridSearch was the most helpful.

The best recall we got was **0.66,** from a GridSearched SVC.

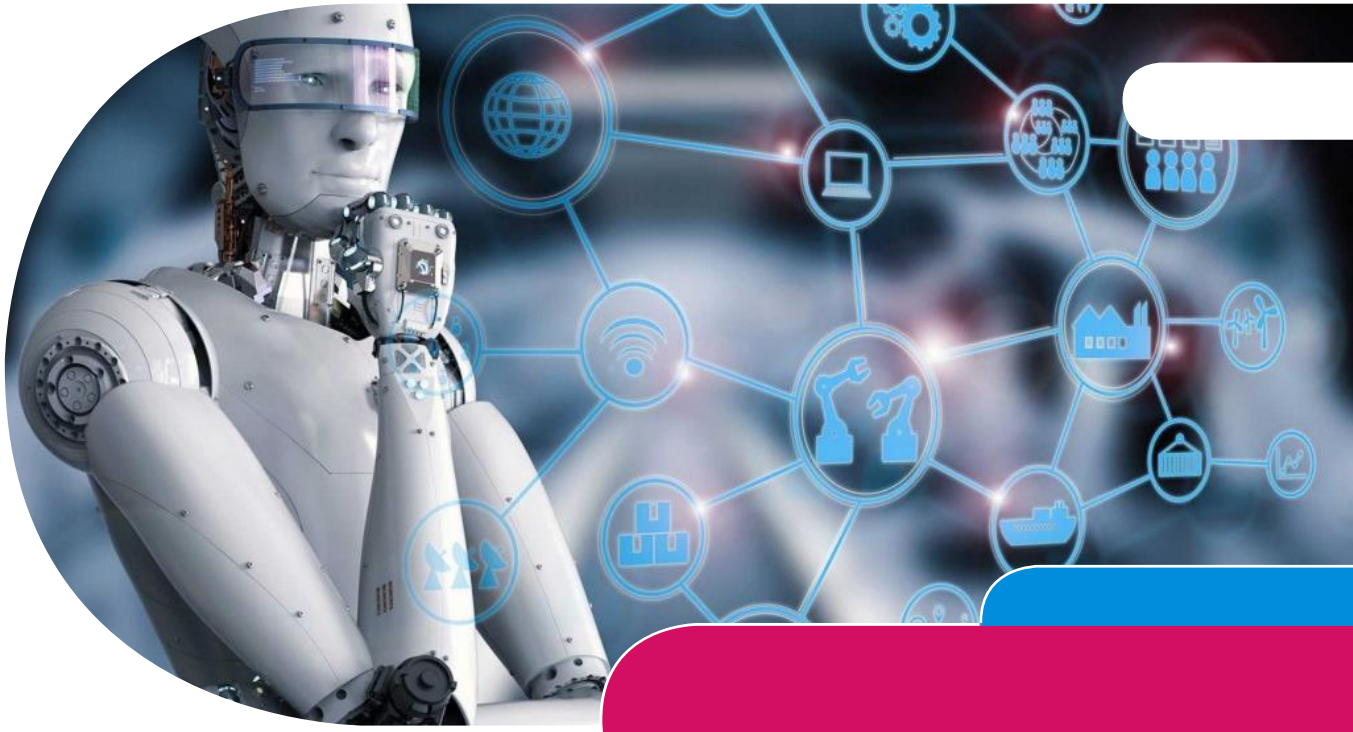


Conclusions

- It is extremely difficult to improve recall without compromising on the other metric heavily.
- Logistic regression and Support Vector Classifiers work the best.
- GridSearch improves performance a lot, but is very slow. Random GridSearch should be looked into.
- It is easy to achieve high accuracy but it doesn’t mean anything if the data itself is imbalanced.

Future Scope

- This research can also be expanded to other datasets available on the internet, such as BayzickBullyingData, TextMiningAndCyberCrimeData, Formspring Data.
- The Wikipedia list of bad words could also be added to the vocabulary to enhance the classification.
- We could also include TF-IDF to get a better result.
- A semi-supervised algorithm like Label Propagation or Label Spreading may also be used.



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THANK YOU!

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